

Quarto Story Telling

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1 Story telling in R

Introduction

These pages introduce time series analysis in R through a recipe-based approach.

Each chapter includes a short conceptual introduction, usually followed by practical, self-contained recipes focused on common tasks.

A version of this textbook is available at <https://nxrolinf.github.io/QuartoSTell>

2 Lab 3: Data Storytelling and Visualization in R

3 Overview

This lab explores data storytelling using R through: - Exploratory data analysis (EDA) - Transformations and skewness - Regression visualization - Time series anomalies - Composite plots - Interactive visualization (Shiny) - A capstone narrative

4 Setup

```
# Install if needed
# install.packages(c("tidyverse", "moderndive", "fivethirtyeight",
#                   "patchwork", "scales", "shiny"))

library(tidyverse)
library(moderndive)
library(fivethirtyeight)
library(patchwork)
library(scales)
```

5 Exercise 1: Exploratory Data Analysis & Skewness

5.1 Raw Distribution

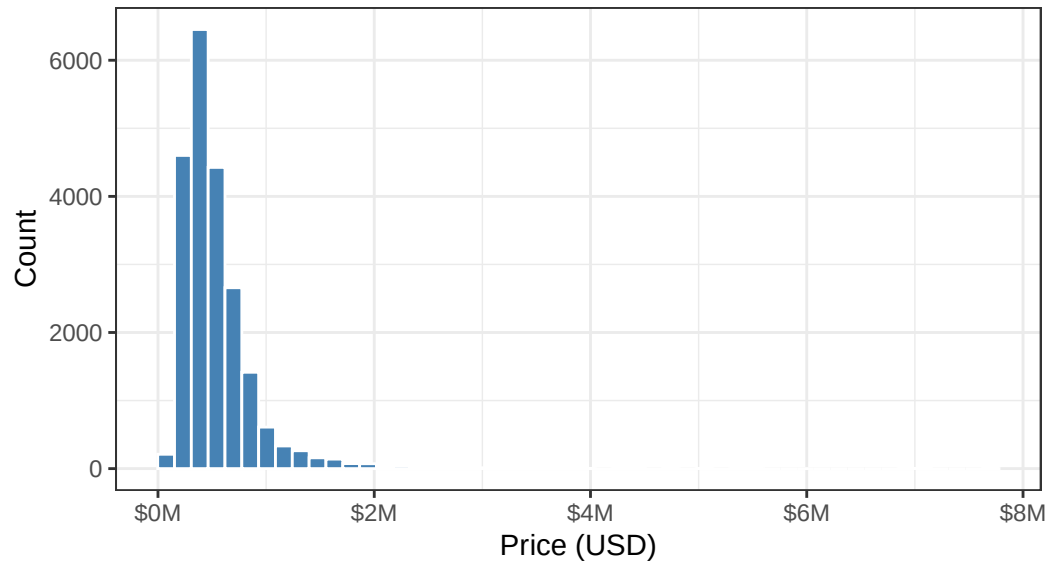
```
data("house_prices")

p1_raw <- ggplot(house_prices, aes(x = price)) +
  geom_histogram(bins = 50, fill = "steelblue", color = "white") +
  scale_x_continuous(labels = label_dollar(scale = 1e-6, suffix = "M")) +
  labs(
    title = "Distribution of House Prices (Raw)",
    subtitle = "Right-skewed distribution",
    x = "Price (USD)",
    y = "Count"
  ) +
  theme_bw()

p1_raw
```

Distribution of House Prices (Raw)

Right-skewed distribution

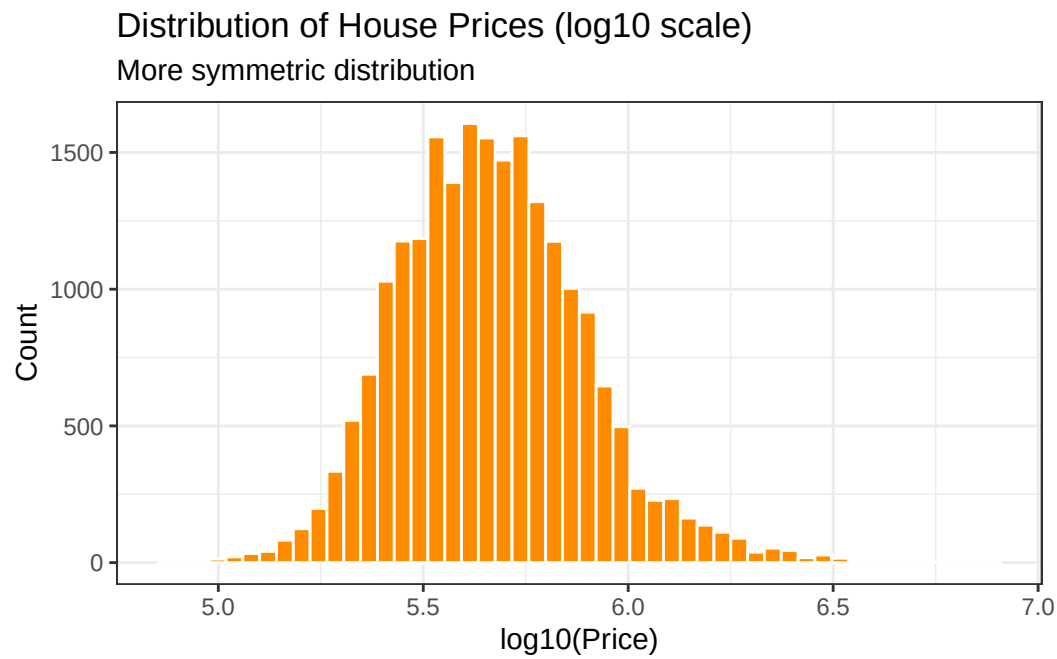


5.2 Log Transformation

```
house_prices <- house_prices %>%
  mutate(
    log10_price = log10(price),
    log10_size = log10(sqft_living)
  )

p1_log <- ggplot(house_prices, aes(x = log10_price)) +
  geom_histogram(bins = 50, fill = "darkorange", color = "white") +
  labs(
    title = "Distribution of House Prices (log10 scale)",
    subtitle = "More symmetric distribution",
    x = "log10(Price)",
    y = "Count"
  ) +
  theme_bw()

p1_log
```



5.2.1 Interpretation

The raw distribution is heavily right-skewed due to high-value outliers.

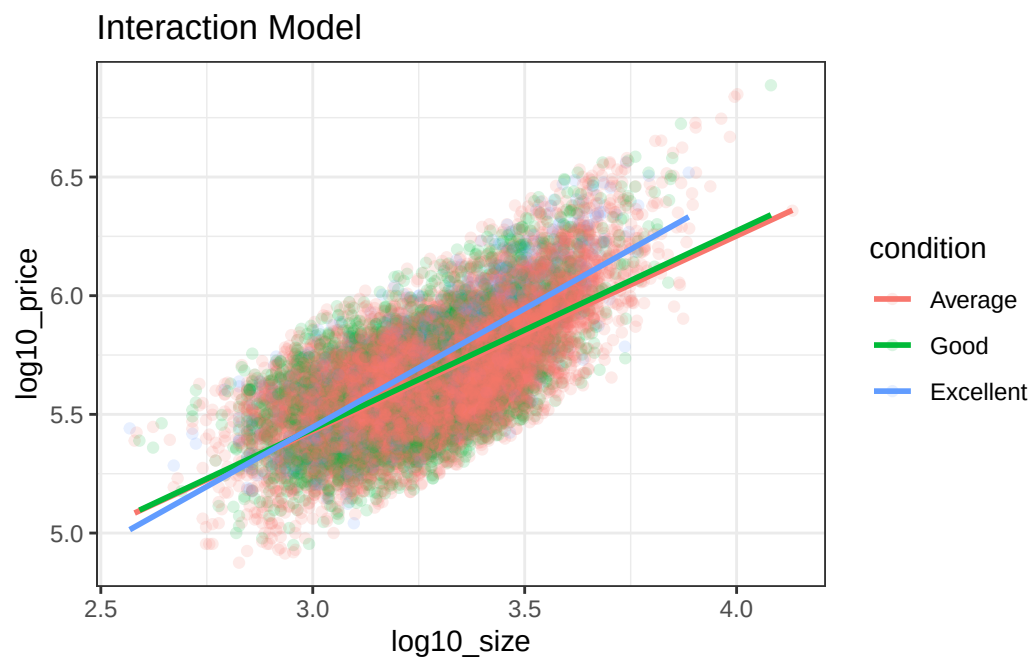
The log transformation reveals a more symmetric, near-normal distribution, improving interpretability.

6 Exercise 2: Interaction vs Parallel Slopes

```
house_prices <- house_prices %>%  
  mutate(condition = factor(condition, levels = 1:5,  
                             labels = c("Poor", "Fair", "Average", "Good", "Excellent")))  
  
house_sub <- house_prices %>%  
  filter(condition %in% c("Average", "Good", "Excellent"))
```

6.1 Interaction Model

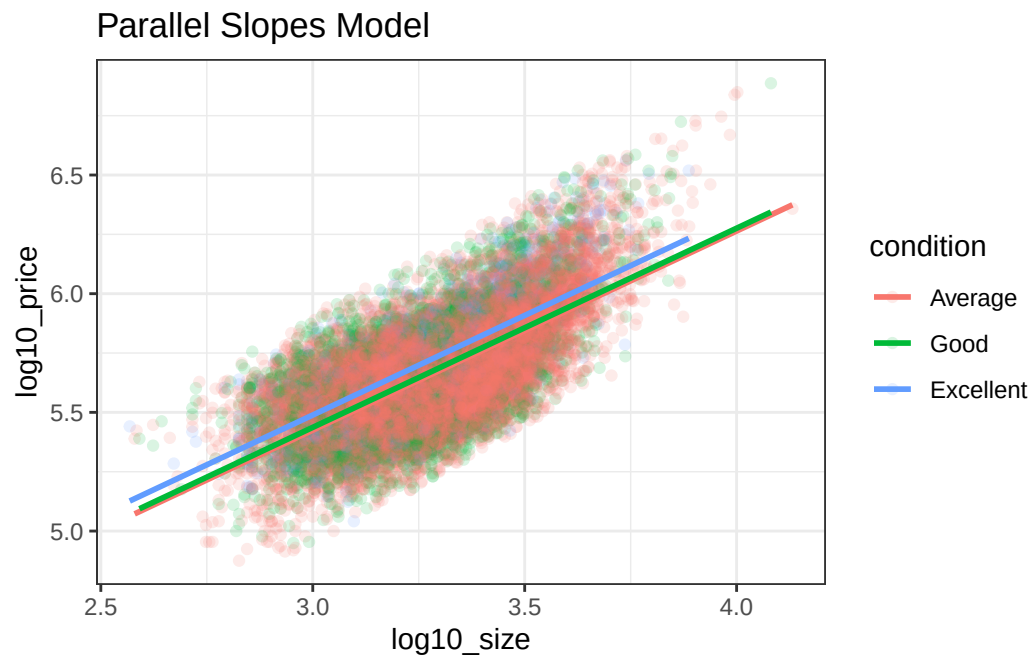
```
p2_interaction <- ggplot(house_sub,  
                         aes(x = log10_size, y = log10_price, color = condition)) +  
  geom_point(alpha = 0.15) +  
  geom_smooth(method = "lm", se = FALSE) +  
  labs(title = "Interaction Model") +  
  theme_bw()  
  
p2_interaction
```



6.2 Parallel Slopes Model

```
p2_parallel <- ggplot(house_sub,
                      aes(x = log10_size, y = log10_price, color = condition)) +
  geom_point(alpha = 0.15) +
  moderndive::geom_parallel_slopes(se = FALSE) +
  labs(title = "Parallel Slopes Model") +
  theme_bw()

p2_parallel
```



6.2.1 Interpretation

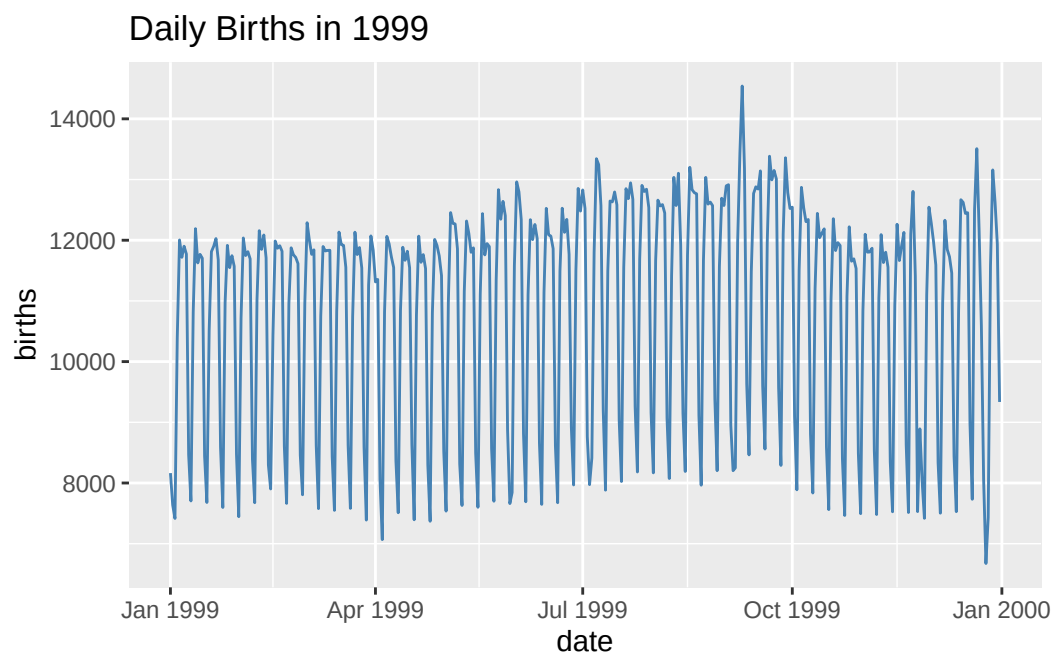
Slopes are similar across conditions, supporting a simpler parallel slopes interpretation: - Size affects price consistently - Condition shifts baseline price

7 Exercise 3: Time Series Anomaly

```
data("US_births_1994_2003")

births_1999 <- US_births_1994_2003 %>%
  filter(year == 1999) %>%
  mutate(date = as.Date(paste(year, month, date_of_month, sep = "-")))

ggplot(births_1999, aes(x = date, y = births)) +
  geom_line(color = "steelblue") +
  labs(title = "Daily Births in 1999")
```



7.1 Peak Detection

```
births_1999 %>%
  arrange(desc(births)) %>%
  head()
```

7 Exercise 3: Time Series Anomaly

```
# A tibble: 6 x 6
  year month date_of_month date      day_of_week births
<int> <int>      <int> <date>      <ord>      <int>
1  1999     9          9 1999-09-09 Thurs      14540
2  1999    12         21 1999-12-21 Tues       13508
3  1999     9          8 1999-09-08 Wed       13437
4  1999     9         21 1999-09-21 Tues       13384
5  1999     9         28 1999-09-28 Tues       13358
6  1999     7          7 1999-07-07 Wed       13343
```

7.1.1 Interpretation

The largest spike occurs on **September 9, 1999 (9/9/99)**, likely due to cultural preferences influencing scheduled births.

8 Exercise 4: Composite Visualization

8.1 Plot A

```
p4a <- ggplot(house_sub,
              aes(x = condition, y = log10_price, fill = condition)) +
  geom_boxplot() +
  theme_bw()
```

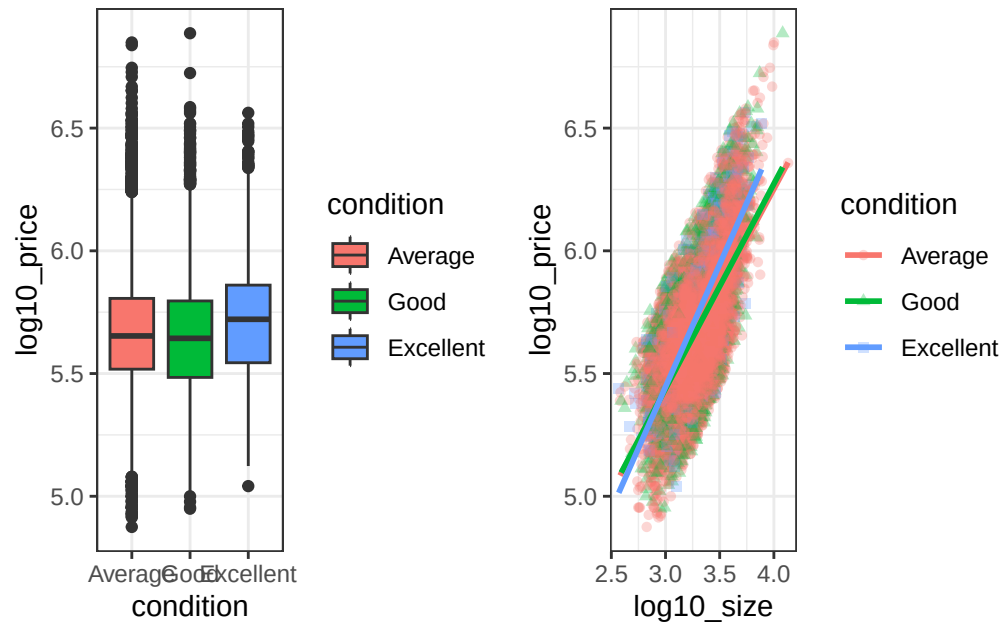
8.2 Plot B

```
p4b <- ggplot(house_sub,
              aes(x = log10_size, y = log10_price,
                  color = condition, shape = condition)) +
  geom_point(alpha = 0.3) +
  geom_smooth(method = "lm", se = FALSE) +
  theme_bw()
```

8.3 Combined

```
p4a | p4b
```

8 Exercise 4: Composite Visualization



9 Exercise 5: Shiny App

Run separately as `app.R`

```
# shinyApp(ui, server)
```

(App code omitted in rendering but included in source file)

10 Capstone: What Drives House Prices?

10.1 Step 1: Distribution

```
cap1 <- p1_raw / p1_log  
cap1
```

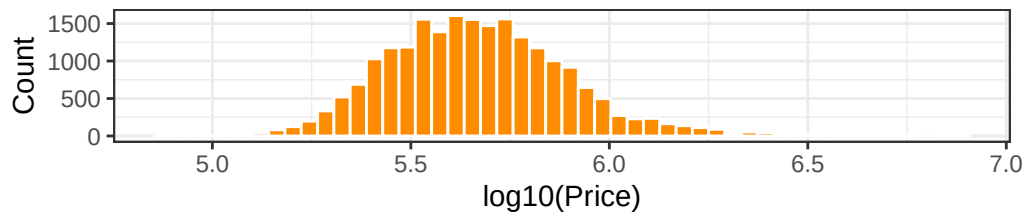
Distribution of House Prices (Raw)

Right-skewed distribution



Distribution of House Prices (log10 scale)

More symmetric distribution

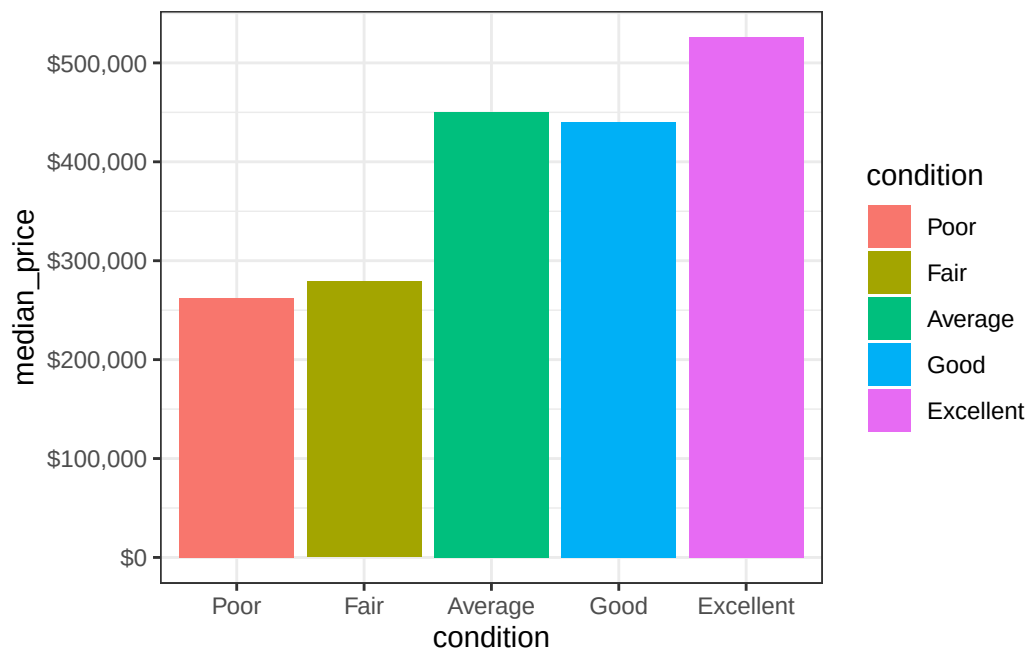


10.2 Step 2: Condition Premium

```
median_by_cond <- house_prices %>%  
  group_by(condition) %>%  
  summarise(median_price = median(price), .groups = "drop")  
  
cap2 <- ggplot(median_by_cond,  
               aes(x = condition, y = median_price, fill = condition)) +  
  geom_col() +
```

```
scale_y_continuous(labels = label_dollar()) +  
theme_bw()
```

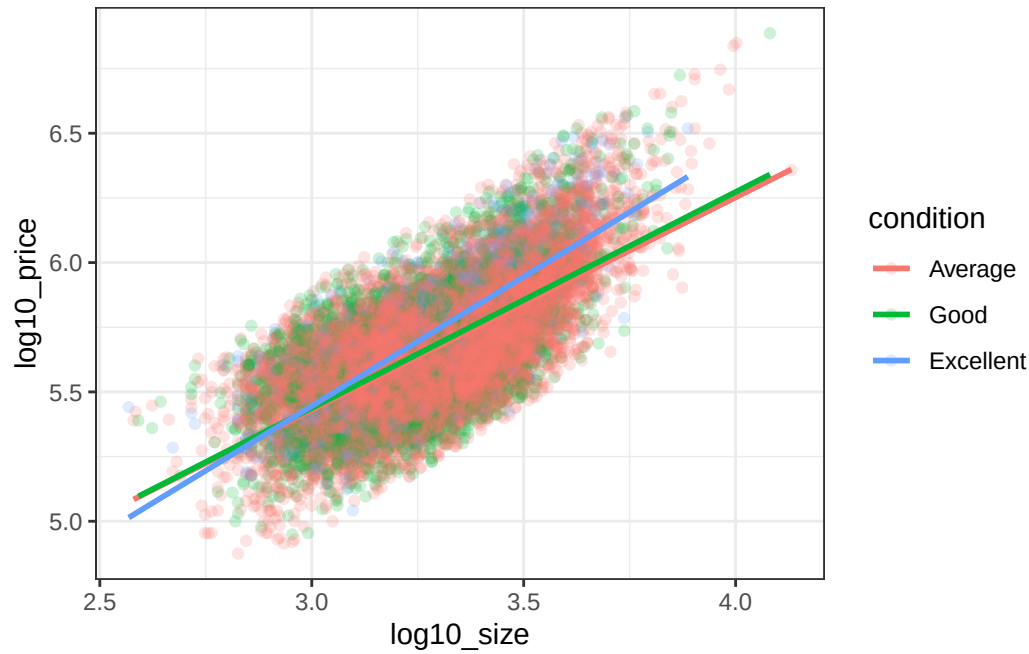
cap2



10.3 Step 3: Size + Condition

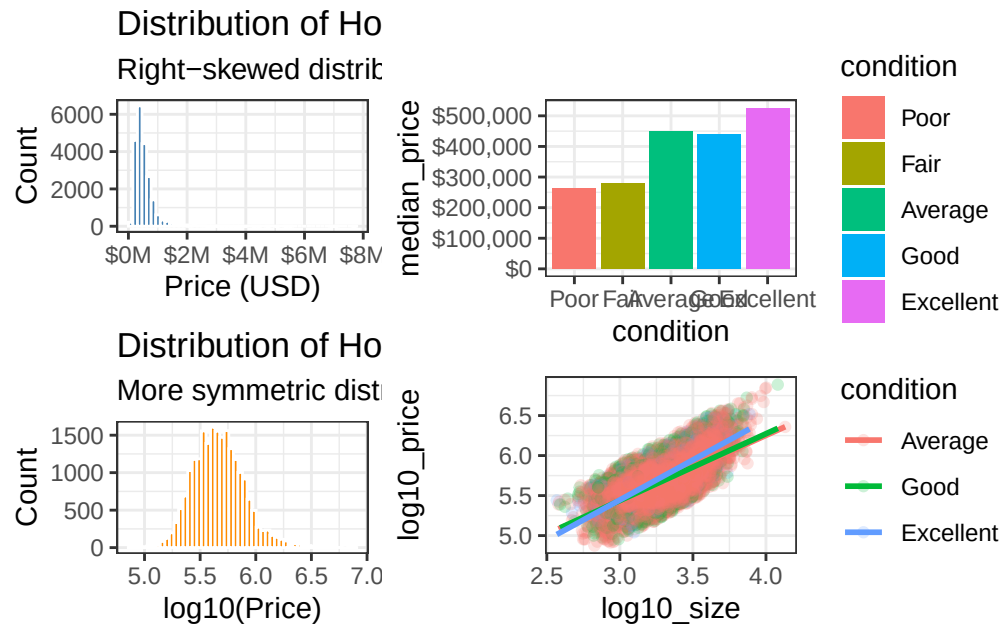
```
cap3 <- ggplot(house_sub,  
               aes(x = log10_size, y = log10_price, color = condition)) +  
  geom_point(alpha = 0.2) +  
  geom_smooth(method = "lm", se = FALSE) +  
  theme_bw()
```

cap3



10.4 Final Narrative

cap1 | (cap2 / cap3)



11 Conclusions

- House prices are **right-skewed**, requiring transformation.
- **Condition strongly affects price levels.**
- **Size increases price consistently across conditions.**

11.1 Call to Action

- Sellers should invest in condition improvements.
 - Buyers can find value in larger homes needing upgrades.
-

12 Appendix: Shiny App Code

```
library(shiny)

house_shiny <- house_prices %>%
  filter(condition %in% c("Average", "Good", "Excellent"))

ui <- fluidPage(
  titlePanel("House Prices Explorer"),
  sidebarLayout(
    sidebarPanel(
      selectInput("condition_select", "Condition:",
                  choices = c("Average", "Good", "Excellent"))
    ),
    mainPanel(plotOutput("scatter_plot"))
  )
)

server <- function(input, output) {
  filtered_data <- reactive({
    house_shiny %>% filter(condition == input$condition_select)
  })

  output$scatter_plot <- renderPlot({
    ggplot(filtered_data(),
            aes(x = log10_size, y = log10_price)) +
    geom_point(alpha = 0.3) +
    geom_smooth(method = "lm") +
    theme_bw()
  })
}

# shinyApp(ui, server)
```


! Work task

For the final part of the lab, you must produce a coherent data story that includes:

1. **At least three different plot types.**
2. **Proper annotation** (titles, axis labels with units, and captions).
3. **A call to action** or a concluding summary of what the visuals reveal.